CHEMOVATE 2.0

TITLE PAGE

- **Problem Statement ID** - CMV26307
- **Problem Statement Title** - AI-Based Personalized Crop Recommendation and Advisory System for Farmers
- **Theme** - AI & Smart Agriculture for Sustainable Farming
- **PS Category** - Software
- **Team Name** - DeepVision

AgroGuard AI

Smart Autonomous Farming Advisor

Proposed Solution

- AgroGuard AI is an intelligent farming support system that integrates Artificial Intelligence, IoT sensors, weather forecasting and satellite soil data to help farmers make better agricultural decisions.
- The system collects real-time field data such as soil moisture, temperature, humidity, and climate conditions to generate personalized crop recommendations and farming guidance.

How It Addresses the Problem

- Provides AI-based crop recommendations based on soil and climate conditions
- Uses IoT sensors to monitor soil moisture and environmental parameters
- Detects plant diseases through image-based AI analysis
- Gives real-time weather alerts and irrigation suggestions
- Predicts crop yield and profitability for better planning

Innovation & Uniqueness

- Integration of AI + IoT + Satellite Data in one platform
- Real-time smart farming advisory system
- AI-based plant disease detection using image analysis
- Sustainability scoring to encourage eco-friendly farming practices
- Supports data-driven decision making for farmers

TECHNICAL APPROACH / METHODOLOGY

Technologies to be Used

- Python (AI/ML models)
- JavaScript

Frontend

- React.js
- Tailwind CSS

Backend

- Node.js

Database

- MongoDB / Firebase

Artificial Intelligence

- Scikit-learn (Crop Recommendation Model)
- TensorFlow / CNN (Plant Disease Detection)

Hardware Components

- ESP32 / Arduino
- Soil Moisture Sensor
- DHT11 Temperature & Humidity Sensor

Implementation Process

Data Collection

IoT sensors collect real-time soil moisture, temperature, and humidity data.

Data Integration

Sensor data is combined with weather forecasts and crop datasets.

AI Processing

Machine learning models analyze the data to recommend the most suitable crops and predict yield.

Disease Detection

Farmers upload crop images which are analyzed by an AI image recognition model to detect plant diseases.

Smart Recommendation System

The platform generates personalized farming advice including irrigation alerts, crop suggestions, and sustainability insights.

User Interface

Farmers view recommendations through a simple web/mobile dashboard.

Sensor Data → AI Model → Crop Recommendation → Farmer Dashboard

FEASIBILITY AND VIABILITY

Feasibility of the Idea

AgroGuard AI is feasible using low-cost IoT sensors, AI models, and cloud-based platforms. Sensors like soil moisture and temperature sensors can collect real-time field data, while machine learning algorithms analyze this data to recommend suitable crops. The system can be developed using open-source tools and APIs, making it practical to implement.

Potential Challenges and Risks

Some challenges include sensor accuracy, limited internet connectivity in rural areas, and availability of reliable agricultural datasets. Environmental conditions may affect sensor readings, and farmers may initially face difficulty adopting digital technologies. These factors could influence the system's performance and usability.

Strategies to Overcome Challenges

To address these issues, the system can use sensor calibration and data filtering to improve accuracy. Offline data storage with periodic synchronization can help handle connectivity issues. Additionally, a simple user interface with multilingual voice assistance can make the platform more accessible for farmers.

IMPACT AND BENEFITS

Potential Impact on the Target Audience

AgroGuard AI empowers farmers with data-driven agricultural insights by combining AI, IoT sensors, and weather data. The system helps farmers make better crop decisions, detect plant diseases early, and manage irrigation efficiently, ultimately improving productivity and reducing farming risks.

Social Benefits

- Improves farmers' knowledge through smart agricultural recommendations
- Reduces dependency on traditional guess-based farming methods

Economic Benefits

- Increases crop yield and farm profitability
- Reduces losses caused by crop diseases and improper crop selection

Environmental Benefits

- Promotes sustainable farming practices
- Optimizes water and fertilizer usage
- Helps maintain soil health and reduce environmental

RESEARCH AND REFERENCES

- FAO – Food and Agriculture Organization
  https://www.fao.org
  Research on sustainable agriculture and smart farming technologies.
- Kaggle Crop Recommendation Dataset
  https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset
  Dataset used for training machine learning models for crop prediction.
- PlantVillage Dataset – Penn State University
  https://plantvillage.psu.edu
  Open dataset used for plant disease detection using image-based AI.